

# A Hybrid Deep Learning Framework for Traffic Sign Recognition using Model Fusion

Mothukuri Abhinaya

Department of CSE

Vignan's Foundation for Science,

Technology and Research

Guntur, India

abhinayamothukuri@gmail.com

Kanneganti Bhavyaratna

Department of CSE

Vignan's Foundation for Science,

Technology and Research

Guntur, India

bhavyaratna23@gmail.com

Maridu Bhargavi

Department of CSE

Vignan's Foundation for Science,

Technology and Research

Guntur, India

bhargaviformal@gmail.com

**Abstract**—The technology of Automatic Traffic Sign Recognition (ATSR) plays an important role in autonomous driving including Advanced Driver Assistance Systems (ADAS) as well as improving road safety. There have been major advancements on this technology through the use of Convolutional Neural Networks (CNN), but there are still several challenges that exist, such as class imbalance, varying scale, and complexity of driving scenarios. The research presented in this paper outlines a framework designed for recognizing traffic signs utilizing the LISA dataset. To extract Regions of Interest (ROIs), image background noise is removed by extracting bounding boxes, then the resulting ROIs are resized, augmented through data augmentation and class-balanced for generalizability. The proposed framework utilizes multiple CNN architectures such as ResNet18, ResNet50, EfficientNet-B0, and DenseNet121 for classification in addition to the use of multiple fusion methods like average and attention-based fusion to improve performance. Experimental results indicate that the proposed system achieved an overall accuracy of 98% while individual model performance was also improved through the use of fusions.

**Index Terms**—Traffic Sign recognition, Deep Learning, EfficientNet, Model Fusion, ROI Extraction, Intelligent Transportation Systems.

## I. INTRODUCTION

Traffic sign recognition (TSR) systems are a key aspect of intelligent transport systems (ITS), ADAS, autonomous driving and improve road safety. By successfully detecting and interpreting the traffic signs around them, vehicles use TSR systems to make safe and correct driving decisions in the real world. TSR systems are at risk of facing many hurdles in real-world scenarios, including variations in illumination, occlusions, background clutter, scale changes and class imbalance. Over the last few years, CNNs and deep learning have become the dominant methods used for TSR primary due to their ability to automatically learn multiple representations of higher and lower level features from the input image without having to be explicitly told what feature, or features, each traffic sign contains [6]. Several types of CNN architectures have been used extensively for traffic sign classification, including ResNet [5], DenseNet [8] and MobileNet [9], and have demonstrated strong and reliable performance. More recently, the advent of EfficientNet [7] has provided researchers with improved accuracy and efficiency

for traffic sign classification by scaling the network's depth, width and resolution through a process too complex for most researchers to achieve without implementing a model scaling method. While there have been many advancements made in terms of algorithms and architecture for traffic sign recognition, it is not uncommon for single-model methods to show signs of insufficient generalization capability - especially when comparing similar traffic signs visually like STOP and YIELD, or when utilizing imbalanced datasets in the training phase. The aforementioned limitations will subsequently result in the need for the use of more robust methods to properly represent the diverse feature space of each traffic sign.

This research presents a hybrid framework for traffic sign recognition using the LISA dataset [2]. The framework has an initial stage called ROI extraction where ROIs are extracted from the bounding box annotations, so background noise can be mitigated and the focus can be on where to detect traffic signs. The ROIs that were extracted are then resized, augmented, and balanced to provide better generalization and reliability of results. In order to obtain the best result from the classification of traffic signs, multiple methods of CNN architecture are used to extract different forms of features from the same image, ResNet18, ResNet50, EfficientNet-B0, and DenseNet121 will all extract unique features of how a traffic sign looks from the same image. Each of the different CNN architectures operates independently and provides its own output that will be combined using various fusion techniques. It has been established that combining model outputs through ensemble learning techniques will produce the best results when the complementary strengths of multiple models are combined [13], [14]. This study uses two approaches to integrating the outputs: average fusion and an attention-based fusion method, which will enhance the overall classification of the traffic signs as well as the generalization ability.

### A. The main contributions of this paper include:

- An ROI extraction-based preprocessing pipeline that improves the classification of traffic signs.

- Comparison of different CNN structures which can be applied in the traffic sign recognition problem.
- An augmentation and balancing strategy implemented a number of times to minimize the imbalance in the dataset.
- Creation and testing of model fusion algorithms such as attention based fusion in order to enhance accuracy and strength.

## II. LITERATURE SURVEY

The application of deep learning and computer vision technology has been investigated in relation to TSR due to the importance of TSR in intelligent transport systems and autonomous vehicle development. The first works to investigate methods for TSR used conventional machine learning methodology. However, in recent years, improvements in deep learning methodologies such as CNNs have demonstrated superior performance compared to conventional machine learning methods by providing the capability to extract hierarchical and spatial information from images.

Benchmark datasets such as the German Traffic Sign Recognition Benchmark GTSRB [1] and LISA Traffic Sign Dataset [2], which are publicly available datasets used to investigate TSR models have been used in literature to evaluate the performance of multiple models using common datasets. These datasets provide ample opportunities to evaluate TSR models under different real world traffic sign environments due to the large amount of variation associated with the illumination conditions, occlusion effects, and image scaling across all data entries; therefore, these datasets can be utilized to evaluate the robustness of a TSR model across various sources of variation. Using early stage deep learning techniques, such as the multi-column deep neural network proposed by Cireşan et al. [4] which achieved high classification rates illustrates that deep architectures are effective methods for use with TSR tasks.

State of the art CNN architectures have advanced performance on TSR to a great extent. The ResNet with residual connections has made it possible to have deeper networks while also helping to significantly reduce the effects of vanishing gradients. DenseNet improved feature reuse by having dense connections between feature maps. EfficientNet was able to improve the performance of the network with the use of compound scaling in depth, width, and resolution, resulting in a greater number of accurate classifications than networks that require more parameters. Additionally, the MobileNet was created to achieve both computational efficiency and accuracy in processing data with a function, making it an ideal architecture for real time applications. Recent surveys, [3] and [16], provide evidence of the performance benefits of using CNN based methods for image based TSR. While CNN based methods have proven effective at solving this problem, the use of only one model often results in poor generalization capabilities, especially in complex real world environments. Various studies, such as [17], have pointed out the need for models that are robust enough to address these challenges. In addition, researchers have evaluated the use of more advanced CNN based architectures, such as Xception

or Vision Transformers, to improve the ability of an image classification model to leverage a globally contextualized approach. However, many of these models require significantly higher computational power.

Ensemble and fusion techniques have been introduced to overcome the limitations of individual models. This type of ensemble learning combines models to provide better predictive performances and generalize better [13], [14]. Multi-model fusion techniques [15] have been used in the area of image classification to take advantage of complimentary features from different architectures and provide significant improvements. Additionally, there are many studies [19] which show that fusion methods based on deep learning provide greater robustness and lower classification error than individual models. Model performance can be enhanced with attention mechanisms in conjunction with fusion approaches. The pioneer Transformer architecture developed by Vaswani et al. [10] laid the foundation for attention mechanisms, which help a model to pay attention dynamically to the most relevant features. Attention modules, such as Convolutional Block Attention Modules (CBAM) [11] used in a convolutional neural network architecture, have been shown to increase a model's feature representation by providing more weight to those important spatially and channel-wise. Even though previous systems have made great strides towards advancing the state of the art (SoTA) in TSR, there has not been a great deal of research that deals with the use of adaptive attention based fusion techniques to dynamically weight the contribution of multiple models to one overall output.

## III. PROPOSED METHODOLOGY

### A. Dataset Description

The LISA Traffic Sign Dataset will be used to evaluate the proposed framework because it is a useful and difficult benchmark for evaluating TSR systems. The images come from actual traffic scenes and allow for actual testing of how robust a model will be to different environments in real life. There are a total of approximately 900 Traffic Sign Instances in the dataset and each of these Traffic Sign Instances has been annotated for classification and localization purposes by bounding box coordinates  $(x_1, y_2)$  and class name such as stop, yield, speed limit, pedestrian crossing allowing for a very good level of accuracy when classifying and locating the Traffic Sign Instances in the data set. The class imbalance in the dataset is an issue that provides significant challenges to the generalization capability of the model. Additionally, the lighting conditions within the dataset the presence of glare, shadows, or low lighting and complex background will make it very difficult to detect traffic signs. The difficulties presented by the variation in viewing angle, occlusion, scale and motion blur will add to the complexity of learning how to develop robust and discriminative features by the model. [2].

### B. Data Preprocessing

Another factor of model performance is the preprocessing. The following would take time to train the dataset. steps:

1) *ROI Extraction*: The former is to get traffic signs given in form of annotations and bounding boxes. The ROI is calculated as:

$$I_{ROI} = \text{Crop}(I, x_1, y_1, x_2, y_2) \quad (1)$$

where  $x_1, y_1$  and  $x_2, y_2$  are the coordinates of the bottom-right and top-left of the bounding box. This eliminates the background noises and enhances the emphasis of features.

2) *Image Resizing*: The extracted ROI images are all resized to a fixed image size of  $224 \times 224$  so that they can be compatible with standard deep learning architectures.

3) *Data Augmentation*: Then we apply various data - augmentation techniques such as random horizontal flipping, random rotation, color jitter reducing brightness, contrast, and saturation, and random affine transformation to promote generalization and eliminate overfitting. These augmentings enhance the strength of the model and level of diversity within the datasets.

4) *Dataset Balancing*: To balance the classes, we generate additional augmented samples until each of them contains the same number of pictures.

$$N_{\text{class}} = \max(N_1, N_2, \dots, N_k) \quad (2)$$

where  $N_1, N_2, \dots, N_k$  represent the desired number of classes and offers homogenous learning into categories.

5) *Data Normalization*: Finally, images are normalized by means of the standard ImageNet values of the mean and standard deviation of images. Normalization stabilization improves and hastens training.

### C. Feature Extraction Model Training

Classifying traffic signs is done in deep learning models, such as ResNet-18, ResNet-50, EfficientNet-B0 and DenseNet-121. Such architectures are chosen because they have demonstrated the capability of classification as well as hierarchical feature representations. In every model, the last and fully connected layer will be substituted with a new layer that is based on the number of traffic sign classes. The prediction of the probability distribution across classes is done using a softmax activation function that comes after this layer.

Each model compares the input images on their own and provides output of the class probabilities. The models are used in parallel to represent various feature representations, which are subsequently used in fusion-based methods to enhance the overall performance.

The following gives the modified classification layer as output: output:

$$y = \text{Softmax}(Wx + b) \quad (3)$$

with  $x$  representing the feature vector that is extracted out of the network,  $W$  and  $b$  representing the matrix of weights and the bias involved in the last layer, respectively, and  $y$  representing the probability distribution that is predicted across the classes.

### D. Loss Function and Optimization

The Cross-Entropy Loss is employed to train the models and is commonly applied to the problems of multi-class classification. Loss is a function that is defined as:

$$L = - \sum y \log(\hat{y}) \quad (4)$$

where  $y$  is the ground truth value and  $\hat{y}$  is the probability of prediction. To optimize, the Adam optimizer is used as it is adaptive having an efficient convergence. The learning rate is set to 0.001.

### E. Model Evaluation

The models are going to be evaluated by the standard measures of classification: precision, recall, accuracy, and F1 -score.

The accuracy measure is the total correctness and it is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} \quad (5)$$

where  $TP$  and  $TN$  are true positives and true negatives, respectively, and Total is also a total of the number of samples.

### F. Model Fusion Techniques

Fusion strategies are used to enhance performance by incorporating predictions of various models.

#### 1. Average Fusion

In two the fusion is averaged as:

$$P_{\text{fusion}} = \frac{P_1 + P_2}{2} \quad (6)$$

The approach standardises the predictions, and it reduces the model errors.

#### 2. Attention-Based Fusion

It is a mechanism which gives the model outputs dynamic weights:

$$P_{\text{fusion}} = w_1 P_1 + w_2 P_2 \quad (7)$$

where  $w_1$  and  $w_2$  and are learned attention weights which are constrained by:

$$w_1 + w_2 = 1 \quad (8)$$

#### IV. MODEL ARCHITECTURE

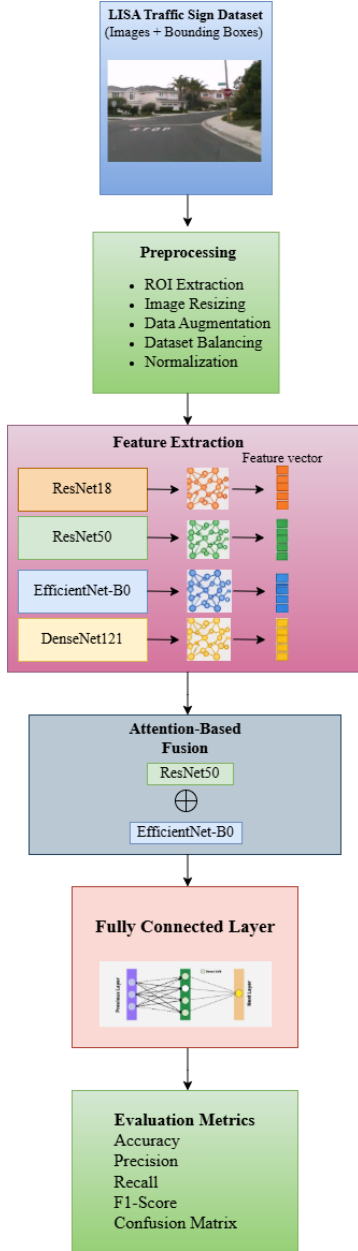


Fig. 1. Proposed Model Architecture

#### V. EXPERIMENTAL RESULTS

##### A. Experimental Setup

The proposed TSR and detection framework was evaluated using the LISA dataset, which consists of 900 annotated images across 9 traffic sign classes. The dataset was divided into 80% training 720 images and 20% testing 180 images. To be classified, images will be resized to  $224 \times 224$  and are trained in a 32 batch during 15 epochs. Here we are using the Adam optimizer with a learning rate of 0.001 and experiments are carried out in a GPU enabled environment.

##### B. Evaluation Metrics

The classification model performance was assessed in terms of the following measures:

- Accuracy
- Precision
- Recall
- F1-score

##### C. Proposed Fusion Model Performance

In order to enhance the classification factor, we employed fusion methods, Average Fusion and Attention based Fusion, to combine the output of multiple deep-learning models.

1) *Average Fusion Model*: The Average Fusion model combines the output of two models by taking the average of their probability scores, thus decreasing the errors in individual models and improving generalization. Experimental results indicate that using fusion yields a performance improvement over using a single, weaker individual model alone. For example, the combination of EfficientNet-B0 and ResNet50 achieved an accuracy of 97.22% while the use of ResNet18 and DenseNet121 achieved 97.78%. Improvements were limited, however; the average fusion model used fixed weights and equal weightings within the process of fusion. The average fusion confusion matrices indicate that most samples in each class were appropriately categorized, as shown by the notably dominant diagonal. Rates of misclassification between visually similar classes speedLimit 25 and speedLimit 35 were reduced when compared to the same class using non-fusion models. Both combinations of average-fusion models produced predictable predictions, meaning that both combinations produced significantly fewer errors and demonstrated predictable behavior. Average fusion produces more stable predictions than either model will alone, thus allowing for reduced variance and an overall improvement to accuracy of predictions resulting from the application of fusion models.

2) *Attention-Based Fusion Model*: The hybrid architecture for image classification incorporates an attention-based fusion method that produces better image classification results than traditional average fusion using model combinations of image classification networks to be published. The results of the proposed method achieve an image classification accuracy of 98.33% when using both models, AFRNet and the hybrid combination of ResNet18 and DenseNet121. This demonstrates that the proposed method utilizes an adaptive weighting mechanism that assigns greater importance to more informative features from each individual model, resulting in improved feature representation capabilities. Consequently, this leads to enhanced overall classification performance due to more effective adaptive and composite weighting in the hybrid image classification models.

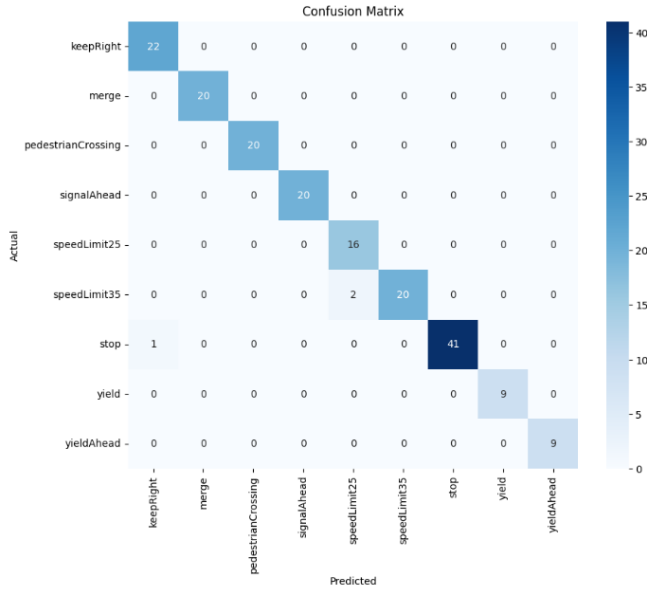


Fig. 2. Confusion Matrix for Proposed AFRNet Model

The AFRNet model has a higher consistency in the predictions as observed in its confusion matrix. The model is more efficient in difficult classes as predictions are stressed out accurately. Misclassifications are also minimised particularly between classes that are visually close. This proves that the attention process enhances the flexibility and stability of the model.

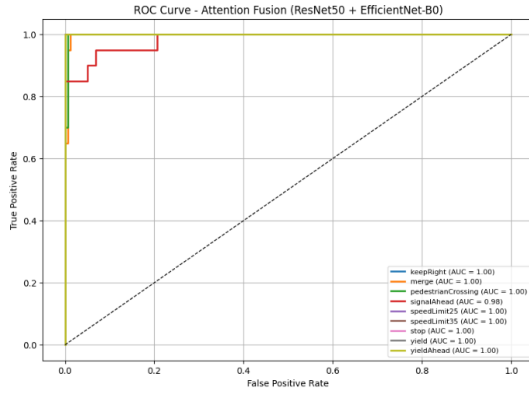


Fig. 3. ROC Curve for Proposed AFRNet Model

In evaluating the classification capability of the attention-based fusion model, the ROC chart compares their True Positive Rate (TPR) to True Positive Rate (FPR) for multiple classes. Because different classes produce various AUCs, the micro AUC provides only a summary of the overall classification effect of the model. Although a few classes achieve fair AUCs, the pattern established indicates that the model can differentiate between the classes of traffic signs. Thus results confirm that this fusion technique is enhancing

the performance of the classification system.

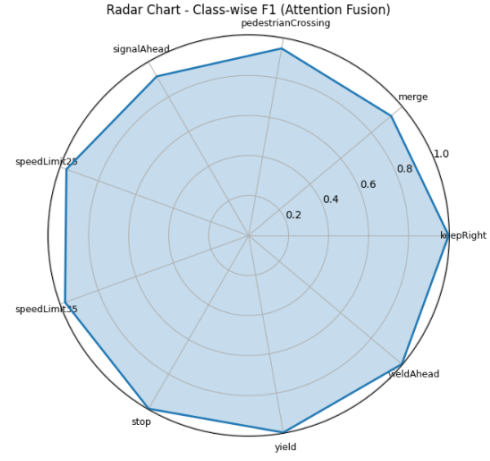


Fig. 4. Radar Chart for Proposed AFRNet Model

Multiple classes of traffic sign has generated F1 scores on this radar plot, where the Attention-based Fusion model produced great results across all classes. Almost every class, including but not limited to Stop, Yield, SpeedLimit35, and KeepRight has generated results that are near perfect F1 scores for precision and recall. The lowest class scores of Merge, PedestrianCrossing, and SignalAhead are most likely the result of challenges with changing lighting, complex backgrounds, and similarities between the visual appearance of some of the classes. The nearly uniform and circular shape of the radar plot indicates that the performance of all of the classes is balanced and that the Attention-based Fusion model has learned to create highly discriminating and accurate traffic sign classification features.

#### D. Classification Performance

TABLE I  
PERFORMANCE COMPARISON OF INDIVIDUAL AND PROPOSED FUSION MODELS

| Model           | Accuracy | Precision | Recall | F1 Score |
|-----------------|----------|-----------|--------|----------|
| ResNet18        | 0.9778   | 0.9795    | 0.9778 | 0.9777   |
| ResNet50        | 0.9500   | 0.9520    | 0.9500 | 0.9503   |
| EfficientNet-B0 | 0.9722   | 0.9744    | 0.9722 | 0.9726   |
| DenseNet121     | 0.9556   | 0.9568    | 0.9556 | 0.9556   |
| AFRNet          | 0.9833   | 0.9848    | 0.9833 | 0.9834   |

Table I summarizes each model's accuracy results. ResNet18 achieved the highest accuracy of all models with 97.78%, closely followed by EfficientNet-B0 which achieved an accuracy of 97.22%. Both models perform very well at successfully extracting relevant features related to traffic signs and identifying them correctly due to their designs. EfficientNet is uniformly scaled which allows it to achieve a balance between

efficient processing and design, whereas ResNet18 utilizes residual connections within its architecture for continual stable learning. DenseNet121 achieved an accuracy of 95.56% which indicates it is able to efficiently reuse features; however, ResNet50's lower accuracy at only 95.00% may indicate that it has been affected by overfitting. In general simple and properly optimized architectures outperform deeper more complicated architectures, thus supporting the notion that efficiency and simplicity should be emphasized instead of complexity in this type of work.

#### E. Fusion Model Performance

TABLE II  
PERFORMANCE COMPARISON OF FUSION MODELS

| Model                          | Accuracy | Precision | Recall | F1 Score |
|--------------------------------|----------|-----------|--------|----------|
| Avg Fusion (R50 + EF-B0)       | 0.9722   | 0.9717    | 0.9721 | 0.9718   |
| Avg Fusion (R18 + DN121)       | 0.9778   | 0.9768    | 0.9774 | 0.9773   |
| Attention Fusion (R18 + DN121) | 0.9833   | 0.9836    | 0.9833 | 0.9833   |
| AFRNet                         | 0.9833   | 0.9848    | 0.9833 | 0.9834   |

Table II shows the performance of the fusion model and compares it to the standalone model. The average accuracy of the entire fusion system was 97.22%, whereas the accuracy of the individual models used in the fusion system was higher at 98.33%. This indicates that there was an improvement in accuracy using the combination of individual baseline models. AFRNet Model performed consistently at over 98.33% across all different combinations of baseline models used to develop the fusion-based platforms, as well as achieving a greater level of precision, recall and F1 scores than the average performance of all of the standalone models. This demonstrates that using adaptive weights of the outputs from each of the individual baseline model contributes significantly to improvement in classification performance under adverse conditions.

#### VI. CONCLUSION

This paper proposes a framework for TSR based on the LISA Dataset that includes preprocessing, several CNNs, and various types of fusion to create a high degree of accuracy and robustness. For example, using ROI extraction via bounding boxes removes background noise and improves upon the classification accuracy of CNNs. Additionally, augmentation and balancing techniques are utilized to create an equal distribution of observations across classes, leading to improved generalization through the reduction of class imbalances. The ResNet18 and EfficientNet-B0 performed best in classification accuracy when comparing individual models, while fusion further enhanced the results by providing moderate increases in accuracy through average fusion and the highest overall accuracy of 98.33% using attention-based fusion. Overall, the proposed framework combines preprocessing, Deep Learning architectures, and fusion techniques in an effective manner that produces

reliable results, making the framework suitable for use in many different applications, including autonomous driving and ADAS.

#### REFERENCES

- [1] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "The German Traffic Sign Recognition Benchmark: A multi-class classification competition," in *Proc. IJCNN*, 2012.
- [2] E. M. Labayrade, J. Douret, and D. Aubert, "LISA Traffic Sign Dataset," University of California, San Diego, 2012.
- [3] Y. Zhu, C. Zhang, D. Zhou, X. Wang, and Y. Bai, "Traffic Sign Recognition Using Deep Learning: A Survey," *IEEE Access*, vol. 8, pp. 202281–202300, 2020.
- [4] J. Cireřan, U. Meier, and J. Schmidhuber, "Multi-column deep neural networks for traffic sign classification," *Neural Networks*, vol. 32, pp. 333–338, 2012.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE CVPR*, 2016.
- [6] C. Szegedy et al., "Rethinking the Inception Architecture for Computer Vision," in *Proc. IEEE CVPR*, 2016.
- [7] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.
- [8] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *Proc. IEEE CVPR*, 2017.
- [9] A. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," arXiv preprint arXiv:1704.04861, 2017.
- [10] A. Vaswani et al., "Attention Is All You Need," in *Proc. NeurIPS*, 2017.
- [11] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module," in *Proc. ECCV*, 2018.
- [12] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *Proc. ICLR*, 2021.
- [13] M. Sagi and L. Rokach, "Ensemble learning: A survey," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 8, no. 4, 2018.
- [14] T. G. Dietterich, "Ensemble methods in machine learning," in *Multiple Classifier Systems*, Springer, 2000, pp. 1–15.
- [15] H. Chen, Y. Zhang, M. Kalantari, and Z. Li, "Multi-model fusion for image classification using deep learning," *IEEE Access*, vol. 7, pp. 146774–146785, 2019.
- [16] X. Lim et al., "Recent advances in traffic sign recognition," *Sensors*, vol. 23, no. 4, 2023.
- [17] Y. Liu, L. Jin, and S. Zhang, "Robust traffic sign recognition using deep convolutional neural networks," *IEEE Transactions on Intelligent Transportation Systems*, 2023.
- [18] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE CVPR*, 2017.
- [19] S. A. Khan, M. Nazir, and A. Khan, "Fusion-based deep learning for image classification: A review," *IEEE Access*, 2021.
- [20] K. Alawaji, "Traffic sign recognition using multi-task deep learning," *Sensors*, vol. 24, no. 11, 2024.